

# Supplementary Material — Individual differences in circadian rhythms measured at a global scale

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This document collects supporting analyses for the main text. Section references, metric names (PC1, PC2,  $\Delta$ MMR) and methods follow the main paper.

## 1 Per-server data summary

The corpus is 92.7 million ranked games from the top ~1,000 players on each of eight regional servers. Servers differ by an order of magnitude in game volume, from 2.3 million games on OC1 to 30.1 million on EUW1 (Table S 1). They also differ sharply in how many players carry an individually significant circadian phase: **JP1 is an extreme outlier**, with roughly three quarters of its players reaching the Benjamini–Hochberg threshold on both behavioural components, against 10–28% everywhere else. This is a symptom of JP1’s daily maintenance shutdown (~05:00–11:00 local; main-text Figure 1): the truncated sampling window makes the fitted 24 h sinusoid look artificially strong and places its peak in unsampled hours. We therefore **exclude JP1 from all phase-based analyses** (the FDR fractions, the modality fits and the kernel density); it is retained only in the period analysis and the play-volume figure. The per-server row for JP1 below is shown for completeness and is flagged accordingly.

Table S 1: Games analysed and the percentage of each server’s ~1,000 players whose PC1 and PC2 peak phase was individually significant (Benjamini–Hochberg FDR < 0.05). JP1 (moved to the foot of the table) is excluded from the pooled phase analyses; its inflated fractions reflect the maintenance-window artifact described above.

| Server         | Ranked games   | PC1 significant | PC2 significant |
|----------------|----------------|-----------------|-----------------|
| BR1            | 10.35 M        | 9.6%            | 5.9%            |
| EUN1           | 13.55 M        | 11.0%           | 6.8%            |
| EUW1           | 30.15 M        | 17.6%           | 20.5%           |
| LA1            | 3.81 M         | 9.1%            | 3.1%            |
| LA2            | 2.91 M         | 9.6%            | 1.5%            |
| NA1            | 17.15 M        | 25.4%           | 28.4%           |
| OC1            | 2.33 M         | 16.9%           | 5.7%            |
| JP1 (excluded) | 12.47 M        | 72.1%           | 76.1%           |
| <b>Total</b>   | <b>92.72 M</b> | —               | —               |

## 2 Within-subject rhythm summary

Each server’s within-subject periodogram peaks within a few minutes of 24.0 h for both behavioural components and for  $\Delta$ MMR (Figure S 1, left; JP1’s two flagged harmonic peaks are the outliers screened from the period meta-analysis): the circadian period is not a pooled artefact but recurs server by server, with the N-weighted mean sitting on 24 h. The fraction of players carrying an individually significant phase separates behaviour from success (Figure S 1, right; JP1 excluded): 14.2% of players for PC1 and 10.3% for PC2, against 0.3% for  $\Delta$ MMR. Skill *change* has almost no per-player circadian signal, which is why it cannot be tested for modality.

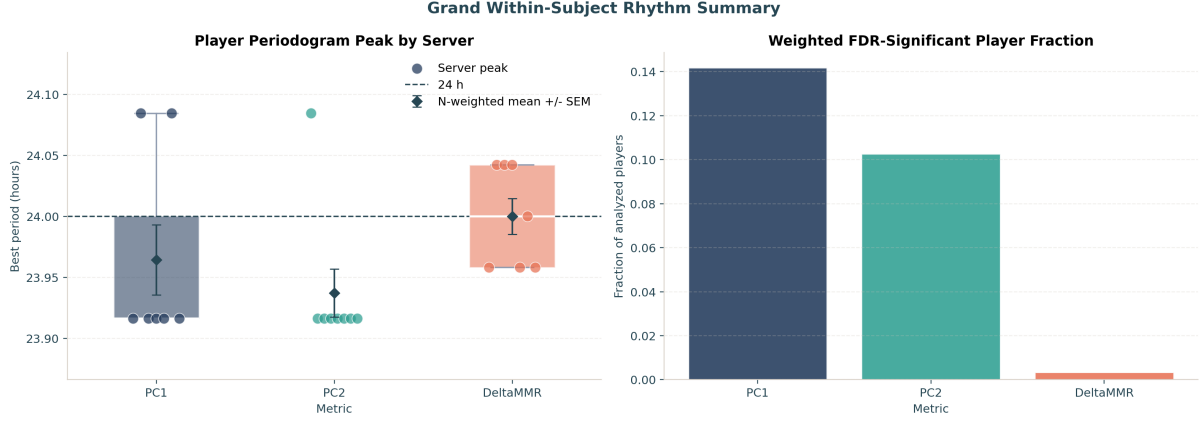


Figure S 1: Within-subject rhythm summary. **Left:** best period from each server’s player-level periodogram for PC1, PC2 and  $\Delta$ MMR; all sit on 24 h (dashed), with the N-weighted mean  $\pm$  SEM overlaid. **Right:** weighted fraction of analysed players whose peak phase was FDR-significant, by metric.

### 3 Phase variation is not inherited from the play-time schedule

The wide individual spread of behavioural peak phases (main text) is not an echo of *when* people play. A natural worry is that the behavioural phases only reproduce the shape of the play-time schedule. We tested this directly. For each analysed player we took the circular-mean hour of their games in local time — their preferred play time — and pooled these across servers (7,966 players), then described the distribution with the same circular fits used for the behavioural phases. This complements Figure 1 of the main text, which makes the point at the level of the *server*: aggregate play volume follows a single evening-peaked curve on every server once local time is used. Here we ask about individual *players*.

The preferred-play-time distribution is single-peaked and comparatively *tight* (Figure S 2, Table S 2): it concentrates in the evening around 19.6 h with a concentration of  $\kappa \approx 2.6$  (circular standard deviation  $\approx 3$  h), and BIC prefers one component over two ( $\Delta$ BIC = 1.8 in favour of the single component). The behavioural peak phases, by contrast, are single-moded but spread more than twice as widely (circular standard deviation  $\approx 7$  h; main text **?@sec-bimodality**). Players cluster their log-on times into the evening far more tightly than their behaviour peaks, so the broad individual variation in behavioural phase is a property of how players behave at a given hour, not of the hours they choose to play.

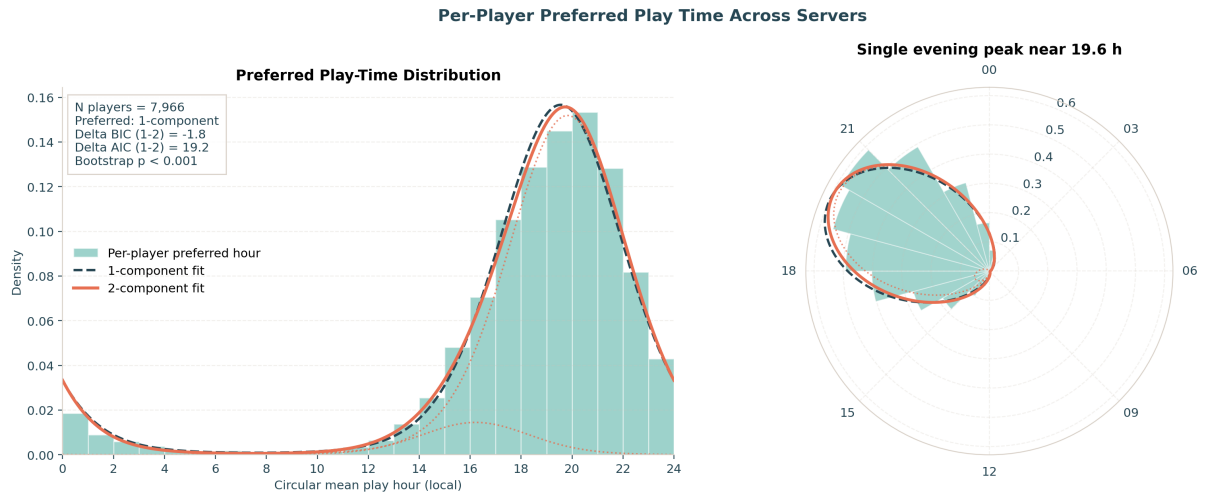


Figure S 2: Per-player preferred play time, pooled across servers ( $N = 7,966$ ). **Left:** distribution of each player’s circular-mean play hour (local), with one-component (dashed) and two-component (solid, dotted components) von Mises fits. **Right:** the same fits in polar view. The distribution has a single evening peak near 19.6 h; BIC prefers one component.

Table S 2: One- versus two-component circular fit to pooled preferred play times. A negative  $\Delta\text{BIC}$  favours the single component: play time is single-peaked, as are the behavioural peak phases in the main text (**?@sec-bimodality**), but the play-time distribution is far more concentrated ( $\kappa \approx 2.6$  vs  $\approx 0.4$ ).

| Metric    | N players | log L (1) | log L (2) | $\Delta\text{AIC}$<br>(1–2) | $\Delta\text{BIC}$<br>(1–2) | Preferred       | Bootstrap<br>$p$ |
|-----------|-----------|-----------|-----------|-----------------------------|-----------------------------|-----------------|------------------|
| Play time | 7,966     | −8,736.8  | −8,724.2  | 19.2                        | −1.8                        | 1-<br>component | <.001            |